Maciej Krzysztof Zuziak

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Profile

AI researcher specialising in the security and privacy of decentralised machine learning – federated learning, membership inference attacks, differential privacy, and data leakage. Marie Skłodowska-Curie Doctoral Fellow (under 10% acceptance rate) and Max Weber Fellow (under 3% acceptance rate) with 5+ years of experience delivering EU-funded research (€20M+ programmes) across academia and industry. Combines rigorous ML engineering with formal legal training in data rights and compliance – able both to design privacy-preserving systems and to assess their regulatory implications. Experienced in independent research leadership, interdisciplinary collaboration, and translating theory into deployable decentralised systems.

Experience

- **Institute of Science Tokyo** — *Oct 2026 – Dec 2026*
  Visiting Researcher (Incoming), School of Computing — Tokyo, Japan
- **European University Institute (EUI)** — *Sep 2026 – Sep 2027*
  Max Weber Fellow, Department of Law — Florence, Italy
- **University of Leeds** — *Jan 2026 – Jun 2026*
  Research Fellow, School of Computer Science — Leeds, United Kingdom
- **National Research Council of Italy (CNR)** — *Nov 2021 – Dec 2025*
  Research Associate, Knowledge Discovery Lab — Pisa, Italy
- **Tellu AS** — *Sep 2023 – Sep 2024 (dual appointment)*
  Visiting Research Engineer, DevOps Department — Oslo, Norway

Education

- **University of Pisa** — *Nov 2021 – Dec 2025*
  Department of Computer Science, Ph.D. (Artificial Intelligence) — Pisa, Italy
- **University of Silesia in Katowice** — *Sep 2016 – Jul 2021*
  Faculty of Law and Administration, Magister Prawa (Master of Laws) — Katowice, Poland

Trainings and Programs

- **FAST (SASH)** — *28 Sep – 2 Oct 2026*
  Frontier AI Security Training – Selected Participant (20 out of 700) — Singapore (remote participation)
- **University of Pisa** — *4–8 Jul 2022*
  Summer School in Artificial Intelligence — Pisa, Italy
- **China University of Political Science and Law** — *Feb 2021 – Jun 2021*
  Academic Exchange Programme — Beijing, China

Journals

First = first author; Co-first = equal contribution, as marked in the publication

[1] **[First]** **M.K. Zuziak**, R. Pellungrini, S. Rinzivillo (2026): **One-Shot Clustering for Federated Learning Under Clustering-Agnostic Assumption**. ACM Transactions on Intelligent Systems and Technology (accepted subject to minor revisions) - ArXiv Preprint.

Conference Papers

First = first author; Co-first = equal contribution, as marked in the publication

[1] **[Co-first]** S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning**. *Proceedings on Privacy Enhancing Technologies* 2026(4), pp. 480–506. Presented at *The 26th Privacy Enhancing Technologies Symposium*, Calgary, Canada - PoPETs Access. (*equal contribution)

[2] J. Nikolic, **M.K. Zuziak**, E. Pournaras (2026): **The Optimization Trilemma: Efficiency, Comfort and Fairness in Decentralized Multi-agent Coordination**. At *IEEE ACSOS 2026*, Cesena, Italy. IEEE (to appear in IEEE Xplore) - ArXiv Preprint.

[3] **[Co-first]** S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini, S. Rinzivillo (2025): **Can Contributing More Put You at a Higher Leakage Risk? The Relationship Between Shapley Value and Training Data Leakage Risks in Federated Learning**. At *SECRYPT 2025*, Bilbao, Spain. SciTePress - SciTePress Access. (*equal contribution)

[4] **[First]** **M.K. Zuziak**, R. Pellungrini, S. Rinzivillo (2024): **One-Shot Clustering for Federated Learning**. At *IEEE Big Data 2024*, Washington, USA. IEEE Xplore - IEEE Xplore Access.

[5] **[First]** **M.K. Zuziak**, S. Rinzivillo (2024): **Amplified Contribution Analysis for Federated Learning**. At *IDA 2024*, Stockholm, Sweden. Springer Lecture Notes in Computer Science (LNCS) - Springer Nature Access.

[6] **[Co-first]** **M.K. Zuziak***, O. Hinrichs*, A. Abdrassulova, S. Rinzivillo (2023): **Data Collaboratives with the Use of Decentralised Learning**. At *ACM FAccT 2023*, Chicago, USA. ACM - ACM Digital Library Access. (*equal contribution)

Workshops, Posters & Opinions

First = first author; Co-first = equal contribution
W = Workshop paper; P = Poster; O = Opinion piece

[W.2] [First] **M.K. Zuziak**, S.Z. El Mestari (2026): **Personal Data, Generated**. Accepted poster at *NeurIPS 2026 Workshop on Privacy in the Era of Large Opaque Models: Theoretical, Legal and Practical Perspectives*, Paris, France (non-archival) - Workshop Link.

[W.1] [First] **M.K. Zuziak**, S. Rinzivillo (2022): **Federated Learning as an Analytical Framework for Personal Data Management - a Proposition Paper**. Hybrid Human Artificial Intelligence (HHAI) Workshop, Amsterdam, Netherlands. CEUR Proceedings - CEUR WS Link.

[P.3] [Co-first] S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning (Poster Version)**. Privacy Enhancing Technologies Symposium, Canada. (*equal contribution).

[P.2] [First] **M.K. Zuziak**, S. Rinzivillo (2024): **Personalization via Clustering in Federated Learning for Accessible Digital Services**. School of Advanced Studies Sant'Anna at Pisa Internet Festival, Italy.

[P.1] [First] **M.K. Zuziak**, S. Rinzivillo (2022): **Federated Learning for Personal Data Management under European Digital Strategy**. Conference on Data Sciences and EU Regulations, IRIT at Université Toulouse III, France.

[O.1] [First] **M.K. Zuziak** (2024): **How to Collaboratively Use Statistical Models in a Secure Way**. *Opinio Juris in Comparatione*. Opinio Juris Link.

Policy Contributions

[1] ACM ETPC (2026). **Comments in Response to European Commission Call for Evidence Survey on the "Digital Omnibus on AI Regulation" Proposal**. Contributors: G. Besanson, M.K. Zuziak, A. Saucedo, F. Medeiros, T. Romanoff. ACM ETPC Link.

[2] ACM ETPC (2025). **Consultation on the Guidelines on Protection of Minors Online under the Digital Services Act**. Contributors: T. Romanoff, A. Saucedo, M.K. Zuziak, M. Beaudouin-Lafon, C.L. Hankin. ACM ETPC Link.

[3] ACM ETPC (2024). **Comments on Draft Delegated Regulation on Data Access (DSA)**. Contributors: A. Deshpande, L. DiBiasi, M.K. Zuziak, J. Rozanec, A. Nagy, A. Saucedo, F. Medeiros, J. Mizeria-Petraszko. ACM ETPC Link.

Recent Talks and Seminars

[1] Institute of Science Tokyo (2026): **On the Intrinsic Vulnerabilities in Federated Learning and Corresponding Defence Methods.**

[2] ACM (2026): **Europe's Digital Omnibus: A New Digital Regime or Simplification.** ACM Recording.

Awards

- **Max Weber Fellowship** *2026*
  European University Institute (under 3% acceptance rate)
- **Frontier AI Security Training (FAST)** *2026*
  Singapore AI Safety Hub (SASH) (under 3% acceptance rate)
- **Horizon Europe (Project Tango) Fellowship** *2024*
  National Research Council of Italy (CNR)
- **Marie Skłodowska-Curie Actions Fellowship** *2021*
  European Commission (under 10% acceptance rate)
- **CSC Scholarship** *2021*
  China Scholarship Council
- **Academic Scholarship (Top 5%)** *2017*
  University of Silesia (awarded multiple times 2017–2020)

Memberships

- **ACM Europe Technology Policy Committee (ETPC)** *Dec 2023 – Present*
  Non-voting member

Skills

Core: Federated Learning, Privacy-Preserving ML, Explainable AI, Data Governance. **Programming:** Python, Julia, C/C++, TypeScript. **ML & Data:** PyTorch, TensorFlow/Keras, JAX, Scikit-Learn, NumPy, Pandas. **Systems:** Flower, PySyft/PyGrid, TensorFlow Federated, Docker, Kubernetes, Linux/WSL. **HPC:** EuroHPC (MeluXina, Leonardo Booster). **LLM Systems:** LangChain, LangGraph, RAG, vector databases (FAISS, Pinecone, Weaviate), LoRA/PEFT.